

BTC-NQ Cross-Market Coherence

A development study of magnitude-conditioned state persistence and incremental Nasdaq futures return value

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Python | scikit-learn | purged walk-forward validation | session-block bootstrap

Finding

Joint move intensity and magnitude balance add a small, repeatable amount of information about next-bar BTC-NQ directional agreement. They do not improve the registered five-minute NQ return forecast over an NQ-only baseline. The state effect also appears overnight, so it is not specific to US cash hours.

Abstract

This study tests whether synchronized, volatility-adjusted Bitcoin and Nasdaq futures candle bodies define persistent cross-market states, and whether relative body magnitude contributes information beyond recent directional coherence. One-minute BTCUSDT perpetual and stitched NQ front-month bars are synchronized in UTC and filtered with DST-aware session calendars. Five-minute bars are primary; one-minute results are mandatory robustness evidence. A nested logistic comparison evaluates state persistence, while matched Ridge models test incremental NQ return value. All preprocessing and regularization selection occur inside purged expanding folds.

Magnitude features reduce primary Brier loss by 0.000699 (95% session-block interval: -0.001120 to -0.000297). In contrast, the cross-market return model increases MSE by 0.1924 bps^2 versus NQ-only (95% interval: 0.0728 to 0.3222), and both Ridge models underperform a fold training-mean benchmark. A prespecified overnight control repeats the state result but not an economic return result. The evidence supports a narrow state-dependence finding, not alpha or deployment readiness.

Primary comparison	Estimate	95% session-block interval	Decision
State Brier: M1 - M0	-0.000699	[-0.001120, -0.000297]	Small development signal
Return MSE: cross - NQ-only	+0.1924 bps^2	[+0.0728, +0.3222]	No incremental value
Overnight state Brier: M1 - M0	-0.000259	[-0.000440, -0.000081]	Not cash-specific

Keywords: cross-market dependence, Bitcoin, Nasdaq futures, calibration, walk-forward validation, data leakage, negative control

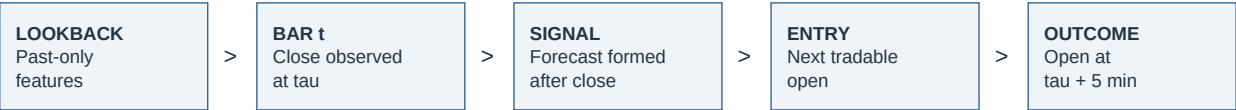
1. Research question and inferential governance

The motivating observation is contemporaneous: BTC and NQ candle bodies often share a direction but differ in volatility-adjusted size. The study does not assume that BTC leads NQ or that the weaker market must catch up. Continuation, reversal and no economically relevant effect remain competing outcomes.

Prespecified questions

Stage	Estimand	Comparison	Role
Descriptive	Signed future NQ return after current agreement	5-minute primary; 15/30/60 secondary	Context
State	Next-bar agreement probability given current agreement	M1 magnitude model minus M0 coherence model	Primary state
Economic	Executable next-five-minute signed NQ return	NQ+BTC Ridge minus NQ-only Ridge	Primary return
Control	Same estimands during 18:00-09:30 ET	Identical definitions and model grids	Negative control

Causal timing



The event bar close is observed before a signal exists. Economic outcomes start at the next tradable open and end at the open five minutes later. No known close is treated as an executable fill.

Multiplicity and closure

The five-minute state and return comparisons are primary for distinct questions. Their 95% complete-session bootstrap intervals are unadjusted. One-minute results, secondary horizons, weekday diagnostics, coefficients and overnight results cannot replace or rescue a primary result. Trial ledgers retain every nested model-selection attempt.

Governance

Protocol v1.0.0 locks a completed development record after results. It is not a preregistration, no final holdout was opened, and no family-wise confirmatory error-control claim is made. New confirmation requires a new protocol and untouched data.

2. Data, sessions and feature construction

Input	Coverage and role	Provenance	Key limitation
BTCUSDT USDT-M perpetual, Binance, 1-minute OHLCV	2024-03-07 to 2026-05-12; synchronized BTC leg	Verified public archive acquisition	Post-ETP only; previously inspected
NQ stitched front-month, CME/IBKR, 1-minute OHLCV	Common range through 2026-05-12; NQ price process	Partial stitched-source provenance	Historical roll rule incomplete; proprietary rows

The deterministic eligibility pass retains **458** complete cash sessions and **551** complete overnight sessions before the 63-session scale warm-up. Entire incomplete or contract-transition sessions are excluded; missing prices are never forward-filled.

Feature definitions

```
body(i,t) = log(close(i,t) / open(i,t))
agreement(t) = sign(body(BTC,t)) * sign(body(NQ,t))
coherence(L,t) = mean agreement over L in {15, 30, 60} minutes
m(i,t) = abs(body(i,t)) / [1.4826 * same-slot MAD over 63 prior sessions]
joint intensity J(t) = sqrt(m(BTC,t) * m(NQ,t))
magnitude balance B(t) = [m(BTC,t) - m(NQ,t)] / [m(BTC,t) + m(NQ,t)]
```

The historical scale is estimated separately by asset, resolution and DST-aware New York session slot. The current session is excluded. Missing history, zero scale, known gaps and roll contamination make an observation ineligible; no epsilon or future-data fallback is allowed.

Parameter provenance

Parameter	Value	Status and rationale
Primary bars	5 minutes	Computationally tractable primary representation; 1-minute mandatory robustness
Coherence clocks	15/30/60 minutes	Used jointly; no best-window selection by outcome
Historical scale	63 sessions	Causal same-slot robust scale; current session excluded
Economic horizon	5 minutes	Next-open to horizon-open; fixed across 1m and 5m representations
Purge	1 full session	Exceeds the five-minute target horizon and prevents adjacent-session leakage
Entry/exit threshold	None	Strategy stage not opened after negative return comparison

3. Purged walk-forward design

All scored predictions are out of fold. The cash-session design starts with 130 feature-ready training sessions, purges one complete session, and scores 12 consecutive 22-session validation blocks. Each outer fold uses three trailing expanding inner validation blocks, again separated by a complete-session purge. The overnight walk is mechanically adapted to the available sample: 135 initial sessions, one purge and 16 blocks of 22.

Nested comparisons

Model	Predictors	Fold-local selection	Primary score
M0 logistic	Coherence 15/30/60 + common direction	Standardization; C in 0.01/0.1/1/10	Brier loss
M1 logistic	M0 + J + B + abs(B)	Independent nested C selection	Brier loss
NQ-only Ridge	NQ direction, magnitude, momentum, time and weekday	Standardization; alpha in 0.01...1000	MSE
Cross-market Ridge	NQ-only + BTC momentum/magnitude + coherence/J/B	Independent nested alpha selection	MSE

Leakage controls

Risk	Control
Look-ahead feature scale	Same-slot MAD uses completed prior sessions only
Preprocessing leakage	Scaler is fitted inside each outer training fold
Hyperparameter leakage	Regularization is selected only through inner purged folds
Outcome overlap and dependence	Complete sessions are purged and bootstrap resampling uses session blocks
Researcher selection	Primary/secondary roles are fixed; all nested attempts enter trial ledgers
Control-result cherry-picking	Overnight control is mandatory and cannot replace the primary

Interpretation rule

A lower loss is better. State evidence requires the upper 95% bound of M1-minus-M0 Brier loss below zero, directionally consistent log loss and no material calibration deterioration. Return evidence requires the upper 95% bound of cross-minus-NQ-only squared-error loss below zero and directionally consistent MAE.

The bootstrap conditions on fitted OOF predictions. It captures clustering within analysis sessions but not uncertainty from rerunning the complete research process on an independent historical sample.

4. Descriptive response and state persistence

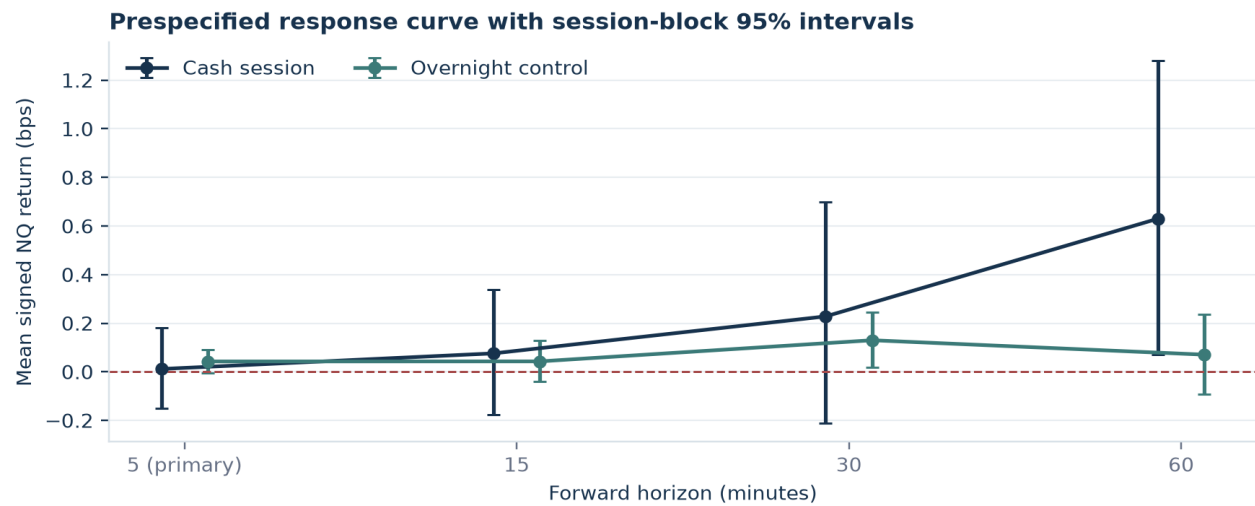


Figure 1. Signed NQ open-to-open response conditional on current BTC-NQ body-direction agreement. Intervals resample complete sessions. The positive 60-minute cash estimate is secondary and cannot replace the five-minute primary.

The cash-session primary descriptive estimate is **0.012 bps** across 19,955 events; its interval includes zero. There is therefore no unconditional five-minute continuation result to serve as a trading premise.

State-model result

Five-minute model	Brier	Log loss	ECE
Training-rate benchmark	0.217250	0.626229	0.009849
M0: coherence	0.216736	0.625038	0.010755
M1: + magnitude	0.216037	0.623370	0.009218

Across 13,724 OOF events, M1-minus-M0 Brier loss is **-0.000699** with interval **[-0.001120, -0.000297]**. The reduction is only about 0.32% relative to M0. It is statistically stable within this development sample, but practically small and not a result about execution or P&L.

5. Incremental return value and overnight control

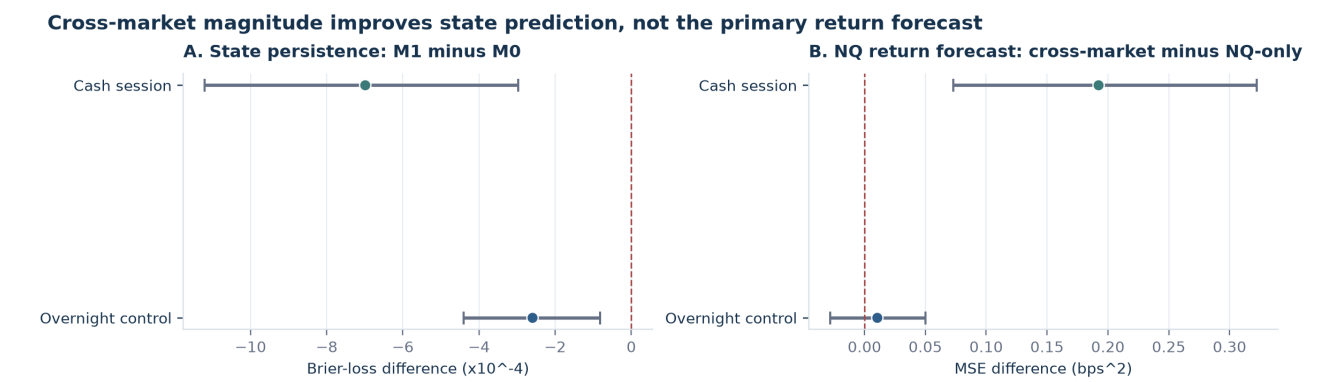


Figure 2. Primary five-minute paired loss differences. Points are OOF means; bars are complete-session bootstrap 95% intervals. Lower favors the expanded cross-market model.

Cash-session return comparison

Model	MSE (bps^2)	RMSE (bps)	MAE (bps)	OOS R2
Fold training mean	134.831	11.612	7.080	0.0000
NQ-only Ridge	135.101	11.623	7.092	-0.0020
Cross-market Ridge	135.294	11.632	7.099	-0.0034

Cross-market-minus-NQ-only MSE is **+0.1924 bps^2** with interval **[+0.0728, +0.3222]**; MAE is also worse. Both Ridge models trail the fold training-mean benchmark. The registered incremental-return criterion is not met.

Overnight negative control

Magnitude also improves overnight state prediction: M1-minus-M0 Brier loss is -0.000259 with interval [-0.000440, -0.000081]. Cash-session specificity is therefore not supported. The overnight return MSE difference is +0.0106 bps^2 with an interval spanning zero, MAE worsens, and both Ridge models again trail the fold mean.

Economic conclusion
The data separate state predictability from return predictability. A small improvement in agreement probabilities does not imply an exploitable NQ return. Costs cannot rescue a gross return effect that is already absent.

6. Calibration, limitations and conclusion

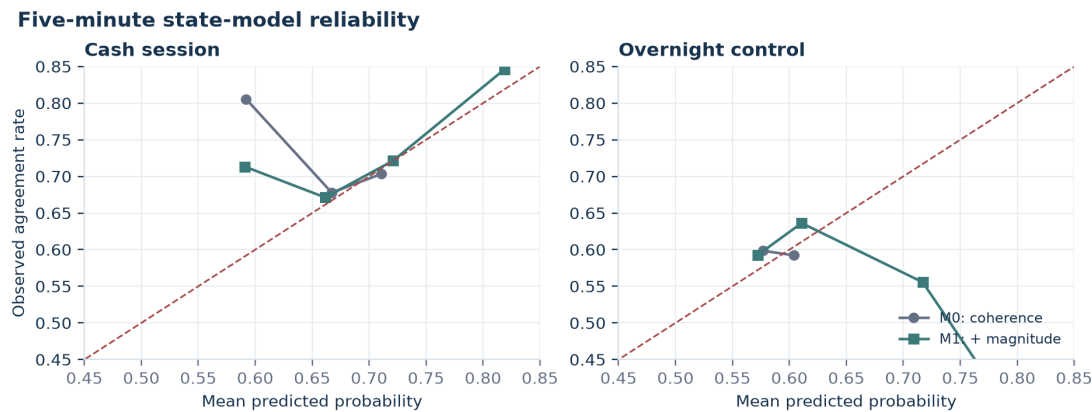


Figure 3. Reliability bins for five-minute M0 and M1 predictions. Empty bins are omitted from the plotted lines but retained in committed tables.

Limitations

Limitation	Consequence
No pre-ETP history	The motivating post-ETF integration narrative is not tested causally or as a before/after design
No pristine holdout	Intervals and model comparisons remain development evidence
Partial NQ provenance	The complete historical roll-selection rule cannot be reconstructed
Systematic missing sessions	Results may not generalize as if session loss were random; weekdays are reported separately
No execution-quality MNQ data	No cost-adjusted strategy, liquidity or deployment claim is made

Conclusion

The study finds a small and reproducible development-sample association between volatility-adjusted joint magnitude and persistence of the BTC-NQ same-direction state. That association repeats in overnight trading, which argues against a cash-session-specific mechanism. The registered linear return test is negative: BTC/coherence variables do not add five-minute NQ return value over an NQ-only baseline. The appropriate conclusion is conditional state dependence without demonstrated alpha.

Reproducibility and credit

All compact results, fold diagnostics, trial ledgers, configurations and SHA-256 manifests are committed in release research-v1.0.0. Proprietary NQ rows remain local. The research question grew from earlier collaborative coursework with Alex Samuseva; this repository is a clean redesign and does not treat legacy notebook results as evidence.

Selected references

Bailey, D. H., Borwein, J. M., Lopez de Prado, M., and Zhu, Q. J. (2014). The Deflated Sharpe Ratio: Correcting for Selection Bias, Backtest Overfitting, and Non-Normality. *Journal of Portfolio Management*.

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